

CSE 314 - Software Engineering Sessional

Online on Behavioral Patterns

Subsection: A1

Time: 30 minutes

A tech company offers a monthly brain-support subscription for patients with severe brain injury, and the service must remain active for the patient to function normally. The subscription has three tiers: **Common**, **Plus**, and **Lux**. Each tier provides a **coverage radius** from regional server towers that determines whether the patient stays responsive during travel: in Common, the safe range is **0–10 km**, while in Plus and Lux it is **0–50 km**. Implement a small simulator with five operations:

- `travelCheck(km)`
- `activateLux(hours)`
- `setMood(calm|exhausted|happy)`
- `promote()` : Takes common to Plus, Plus to Lux, Lux to Lux (no change)
- `demote()` : Takes Lux to Plus, Plus to Common, Common to Common (no change)

When `travelCheck(km)` is called, the system should print whether the patient is **STABLE** or **UNSTABLE** at that distance under the current tier; if the distance exceeds the tier's safe range, the patient becomes **UNSTABLE** (the patient blacks out), and the log should print alert message which asks to bring the patient back into coverage. When in coverage after which the simulation continues by calling the `travelCheck(0km)` and the patient regains consciousness.

The `activateLux(hours)` operation should temporarily switch the service into Lux for the specified duration (you may use *sleep()*), then automatically return to whichever tier (Common or Plus) was active right before Lux was activated. Finally, `setMood(...)` should only work while Lux is active; otherwise it prints "Mood control unavailable."

Demonstrate your design using print logs showing a few distance checks, some promotion-demotions, one Lux activation and return to the earlier tier and mood changes.

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Subsection: A2

Time: 20 minutes

King's Landing has a giant message board where ravens deliver scrolls like:

- "Enemy spotted near the river"
- "Winter supplies running low"
- "Ships seen in the east"

Your task is to build a RavenBoard system with appropriate design pattern where multiple groups can receive new messages: Commander, Scouts, Supply Team etc. and act based on it. (example: Scouts print "Dispatch riders!", Supply Team prints "Update inventory!"). However, groups can subscribe/unsubscribe at runtime (like Scouts leaving the board room).

Demonstrate your implementation using 3 messages, some subscribe/unsubscribe and some response prints.

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Subsection: B2

Time: 20 minutes

You are developing a Smart Home Automation Hub. You have various devices: a Light Sensor, Automatic Blinds, and an Air Conditioner.

To keep the system organized, the devices should not talk to each other directly. Instead, they all report to the Central Hub.

- When the Light Sensor detects “High Brightness,” it notifies the Hub. The Hub then tells the Blinds to close.
- When the Blinds close, they notify the Hub. The Hub then tells the Air Conditioner to turn on because the room will get stuffy.

Task: Choose the appropriate design pattern to solve this problem and implement a minimal demonstration.

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Subsection: C1

Time: 25 minutes

You are building a small hospital visit simulator. Every patient visit must follow the same overall flow, and the system should print each step in order. The hospital has different departments, and each department follows the same visit structure, but customizes a few steps.

1. Check-In (register patient name + assign visit ID)
2. Record Vitals (print temperature + blood pressure)
3. Assessment
4. Treatment
5. Discharge Summary (print “patient discharged” + notes)

General Department

Assessment: “Doctor performs normal diagnosis”

Treatment: “Prescribe standard medicine”

Pediatrics Department

Assessment: “Doctor checks symptoms by ensuring child comfort level”

Treatment: “Give child-safe medicine, friendly reassurance message”

Emergency Department

Assessment: “Quick triage check (urgent/non-urgent)”

Treatment: “Immediate emergency procedure”

Task: Implement the treatment system with appropriate design pattern and run the simulation for 1 patient in each department, showing the printed flow clearly.

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Subsection: C2

Time: 25 minutes

You are building a financial dashboard for a day-trading firm. The traders need to see real-time changes in stock prices across various distinct widgets on their screen.

The core of the system is a StockData feed. Whenever the price of a specific stock changes, several unrelated components need to update immediately:

4. A **Ticker Tape** widget needs to scroll the new price.
5. A **Graph** widget needs to plot the new data point.
6. A **Buy/Sell Bot** needs to evaluate if the new price triggers an automated trade.

You need a design where the StockData feed doesn't know the specifics of these widgets but can notify all of them automatically whenever a price update occurs. Also there should be provision of adding or removing widgets.

Task: Choose the appropriate design pattern to solve this problem and demonstrate with minimal code.